

STEP 1: Create Project Folder

```
mkdir program-1  
cd program-1
```

```
1RV24MC087_SSDhanush@dhanush-Latitude:~$ mkdir program-1  
cd program-1
```

STEP 2: Create Dockerfile

```
nano Dockerfile
```

```
1RV24MC087_SSDhanush@dhanush-Latitude:~/program-1$ cat Dockerfile  
# Use a slim Python base image  
FROM python:3.9-slim  
  
# Set working directory  
WORKDIR /app  
  
# Copy requirements file and install dependencies  
COPY requirements.txt .  
RUN pip install --no-cache-dir -r requirements.txt  
  
# Copy application code  
COPY . .  
  
# Set Flask environment variables  
ENV FLASK_APP=app.py  
ENV FLASK_RUN_HOST=0.0.0.0  
ENV FLASK_RUN_PORT=5000  
  
# Expose port  
EXPOSE 5000  
  
# Run Flask app  
CMD ["flask", "run"]
```

STEP 3: Create app.py

```
nano app.py
```

```
1RV24MC087_SSDhanush@dhanush-Latitude:~/program-1$ cat app.py  
from flask import Flask  
  
app = Flask(__name__)  
  
@app.route("/")  
def hello():  
    return "Hello, Docker!"  
  
if __name__ == "__main__":  
    app.run(host="0.0.0.0", port=5000)
```

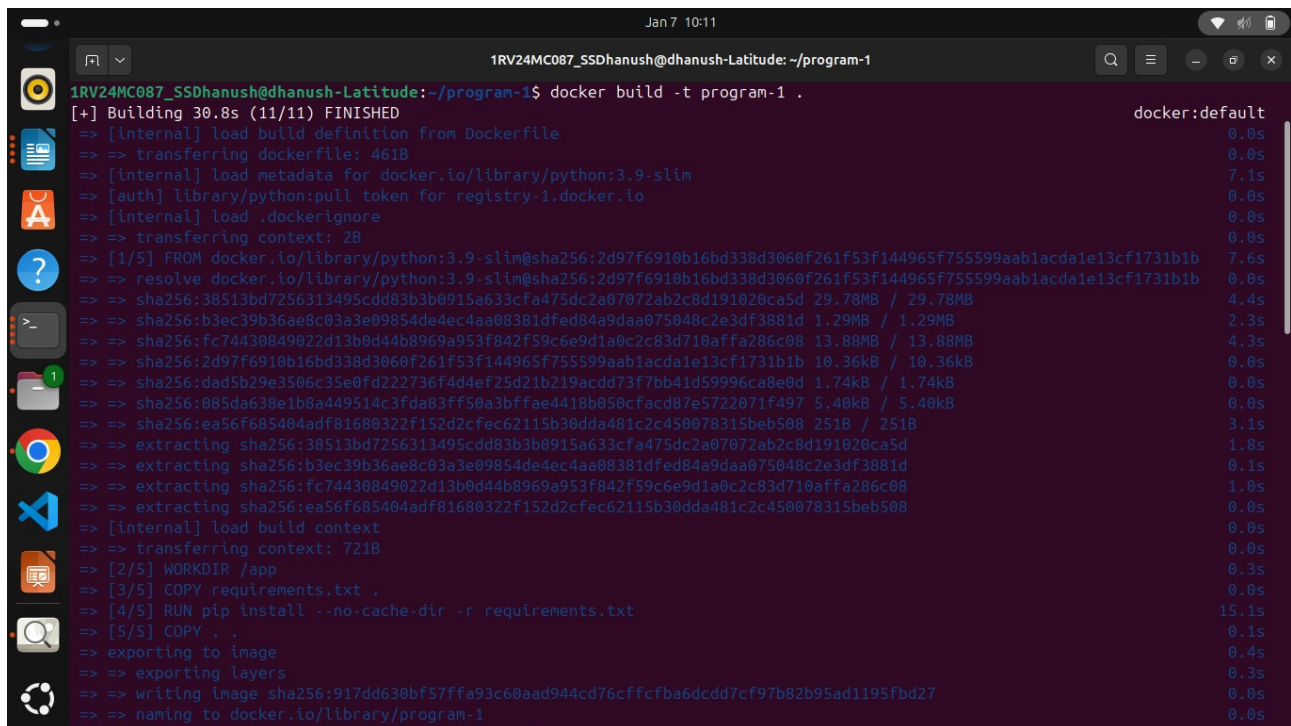
STEP 4: Create requirements.txt

```
nano requirements.txt
```

```
1RV24MC087_SSDhanush@dhanush-Latitude:~/program-1$ cat requirements.txt  
flask  
1RV24MC087_SSDhanush@dhanush-Latitude:~/program-1$
```

STEP 5: Build Docker Image

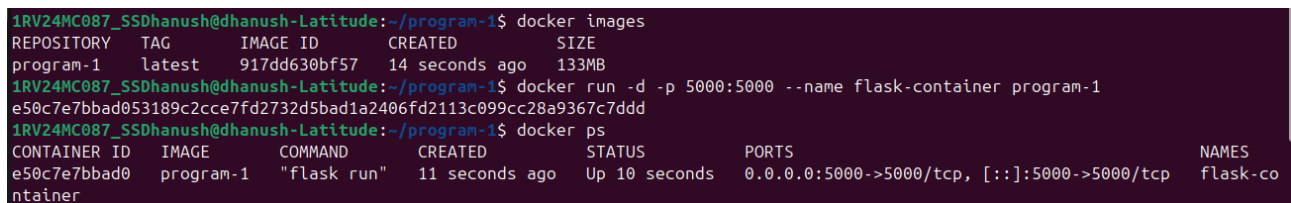
`docker build -t program-1 .`



```
1RV24MC087_SSDhanush@dhanush-Latitude: ~/program-1
1RV24MC087_SSDhanush@dhanush-Latitude:~/program-1$ docker build -t program-1 .
[+] Building 30.8s (11/11) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 461B
=> [internal] load metadata for docker.io/library/python:3.9-slim
=> [auth] library/python:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/5] FROM docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acda1e13cf1731b1b
=> => resolve docker.io/library/python:3.9-slim@sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acda1e13cf1731b1b
=> sha256:38513bd7256313495cdd83b3b0915a633cfa475dc2a07072ab2c8d191020ca5d 29.78MB / 29.78MB
=> sha256:b3ec39b36ae8c03a3e09854de4ec4aa08381dfed84a9daa075048c2e3df3881d 1.29MB / 1.29MB
=> sha256:fc74430849022d13b0d44b8969a953f842f59c6e9d1a0c2c83d710affa286c08 13.88MB / 13.88MB
=> sha256:2d97f6910b16bd338d3060f261f53f144965f755599aab1acda1e13cf1731b1b 10.36kB / 10.36kB
=> sha256:dad5b29e3506c35e0fd222736f4d4ef25d21b219acdd73f7bb41d59996ca8e0d 1.74kB / 1.74kB
=> sha256:085da638e1b8a449514c3fda83ff50a3bffa4418b050cfacd87e5722071f497 5.40kB / 5.40kB
=> sha256:ea56f685404adf81680322f152d2cfec62115b30dda481c2c450078315beb508 251B / 251B
=> extracting sha256:38513bd7256313495cdd83b3b0915a633cfa475dc2a07072ab2c8d191020ca5d
=> extracting sha256:b3ec39b36ae8c03a3e09854de4ec4aa08381dfed84a9daa075048c2e3df3881d
=> extracting sha256:fc74430849022d13b0d44b8969a953f842f59c6e9d1a0c2c83d710affa286c08
=> extracting sha256:ea56f685404adf81680322f152d2cfec62115b30dda481c2c450078315beb508
=> [internal] load build context
=> => transferring context: 721B
=> [2/5] WORKDIR /app
=> [3/5] COPY requirements.txt .
=> [4/5] RUN pip install --no-cache-dir -r requirements.txt
=> [5/5] COPY . .
=> exporting to image
=> exporting layers
=> writing image sha256:917dd630bf57ffa93c60aad944cd76cfcfba6dcdd7cf97b82b95ad1195fbd27
=> naming to docker.io/library/program-1
```

STEP 6: Run Docker Container

`docker run -d -p 5000:5000 --name flask-container program-1`



```
1RV24MC087_SSDhanush@dhanush-Latitude:~/program-1$ docker images
REPOSITORY TAG IMAGE ID CREATED SIZE
program-1 latest 917dd630bf57 14 seconds ago 133MB
1RV24MC087_SSDhanush@dhanush-Latitude:~/program-1$ docker run -d -p 5000:5000 --name flask-container program-1
e50c7e7bbad053189c2cce7fd2732d5bad1a2406fd2113c099cc28a9367c7ddd
1RV24MC087_SSDhanush@dhanush-Latitude:~/program-1$ docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
e50c7e7bbad0 program-1 "flask run" 11 seconds ago Up 10 seconds 0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp flask-co
ntainer
```

STEP 7: Test in Browser

Open browser



STEP 8: Stop the Container

```
docker stop flask-container
```

STEP 9: Remove Container

```
docker rm flask-container
```

STEP 10: Remove Docker Image

```
docker rmi program-1
```

```
1RV24MC087_SSDhanush@dhanush-Latitude:~/program-1$ docker stop flask-container
flask-container
1RV24MC087_SSDhanush@dhanush-Latitude:~/program-1$ docker rm flask-container
flask-container
1RV24MC087_SSDhanush@dhanush-Latitude:~/program-1$ docker rmi program-1
Untagged: program-1:latest
Deleted: sha256:917dd630bf57ffa93c60aad944cd76cffcfba6dcdd7cf97b82b95ad1195fbd27
```